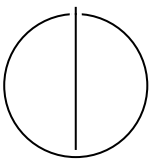
SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Reduction of Memory in Tensor Network Contractions

Fam Shihata



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Speicherreduzierung bei Tensornetzwerkkontraktionen

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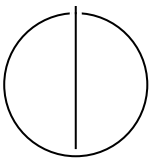
Submission Date:	Submission date



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Munich, Submission date Fam Shihata

Acknowledgments

Abstract

Contents

1 Introduction

Tensor network contraction (TNC) has become a core computational primitive in fields ranging from quantum many-body physics and quantum circuit simulation to chemistry and high-dimensional machine learning and data analysis. Their appeal comes from the ability to represent and manipulate exponentially large objects through structured factorizations, often making previously intractable problems accessible in practice [5]. Yet this promise comes with a major obstacle: the cost of contracting a tensor network can grow dramatically depending on how intermediate tensors are formed. In many realistic settings, the limiting factor is not only the number of arithmetic operations, but the peak memory required to store these intermediates. A contraction strategy that is theoretically efficient in floating-point operations (FLOPS) may still be unusable in practice if it creates tensors too large to fit in memory.

This makes memory reduction a central problem rather than a secondary implementation concern. Prior work on contraction ordering has already shown that the sequence in which tensors are contracted has a decisive impact on computational feasibility and overall performance [17, 18]. At the same time, high-performance tensor contraction frameworks and tensor-transposition libraries have demonstrated that practical efficiency depends heavily on how contractions are mapped onto hardware, how data is rearranged, and how temporary storage is managed [12, 21]. In other words, optimizing tensor networks requires attention not only to algebraic structure, but also to memory footprint, data movement, and layout-aware execution.

The importance of optimizing tensor network contractions originates from both theoretical and practical considerations. From a theoretical perspective, tensor contraction is known to be computationally hard in general, with complexity that can grow exponentially with the size of the network [17]. From a practical standpoint, modern applications such as quantum circuit simulation and many-body physics rely heavily on large-scale tensor contractions, where inefficiencies quickly become restrictive.

A key observation in high-performance computing (HPC) is that memory bandwidth and data movement increasingly dominate execution time, often surpassing the cost of FLOPS [12, 13]. In large-scale simulations, such as those targeting quantum supremacy benchmarks [22, 13], the ability to efficiently manage memory and minimize data movement is critical to achieving feasible runtimes.

Moreover, the peak memory usage during tensor contraction directly determines

whether a computation is even possible on a given system. Many tensor network contractions fail not due to computational limits, but because intermediate tensors exceed available memory. As a result, reducing peak memory usage is not merely an optimization objective, but a fundamental requirement for scalability.

Despite these challenges, most existing optimization techniques primarily focus on reducing computational cost (FLOPS) through contraction ordering, often overlooking memory-related factors such as tensor layout, data locality, and permutation overhead. This creates a gap between theoretical optimization and practical performance, particularly on modern architectures where memory hierarchy plays a central role.

1.1 Problem Formulation and Challenges

Ultimately, the goal of contraction is to compute a final tensor by successively contracting pairs of tensors along shared indices. This process involves three key components: the contraction order, which determines the sequence in which tensor pairs are combined; the intermediate tensor shapes, which define the sizes and dimensionalities of tensors generated during the contraction; and the tensor layout and permutation, which determine how tensor indices are arranged in memory and therefore how efficiently operations can be executed.

While contraction ordering has been extensively studied, the challenges associated with intermediate tensor shapes and tensor layouts remain significant and often underexplored. In particular, tensor permutations play a crucial role in enabling efficient computation. High-performance implementations typically rely on transforming tensor contractions into matrix multiplications via reshaping and permutation, a technique known as Tensor-Tensor Generalized Multiplication (TTGT) [21].

However, these permutations introduce additional memory movement, which can be costly. In many cases, tensors are repeatedly permuted and reshaped throughout the contraction sequence. This leads to increased memory bandwidth usage, poor cache locality, and higher peak memory consumption due to the creation of intermediate copies. These effects collectively degrade performance and can become a dominant bottleneck in large-scale computations.

Furthermore, existing approaches typically treat permutations as local operations required to enable individual contractions. This local perspective ignores the global structure of the contraction sequence and can result in redundant or suboptimal data transformations that accumulate over the course of execution [4].

1.2 Research Aim

This thesis proposes a complementary approach that shifts part of the optimization focus to the *post-contraction phase*. Instead of solely optimizing the contraction order, we investigate how global tensor permutations can be applied after determining a contraction path to improve memory efficiency.

The key idea is to treat tensor layout as a first-class optimization variable. By analyzing the entire contraction sequence, it becomes possible to reduce redundant permutations across consecutive contractions, improve data locality and cache utilization, and minimize peak memory usage by carefully controlling intermediate tensor layouts.

This approach is relatively niche compared to traditional methods, as most existing work focuses on pre-contraction optimization and local permutation strategies. However, given the growing importance of memory efficiency in modern computing systems, global permutation optimization presents a promising direction for further research.

1.3 Contributions of This Work

In summary, this thesis makes several key contributions. It introduces a novel framework for post-contraction optimization based on global tensor permutations. It provides a detailed analysis of the impact of tensor layouts on memory usage and data movement. It highlights the limitations of existing pre-contraction optimization techniques in addressing memory-related challenges. Finally, it demonstrates how tensor permutation strategies can complement contraction ordering to achieve improved overall performance.

By bridging the gap between contraction planning and memory-aware tensor layout optimization, this work aims to provide a more comprehensive approach to efficient tensor network contraction.

2 Background

2.1 Foundations of Tensor Networks

2.1.1 Tensors

For our discussion, it is important to define what tensors are, how they are contracted, and how they are represented in networks. A tensor is a multidimensional array that generalizes the concept of vectors and matrices to arbitrary dimensions. Formally, a tensor of order n (n-mode tensor or rank n tensor) is a collection of entries indexed by n indices, each ranging over a finite set of dimension sizes. For instance, a matrix is a second-order ($n = 2$) tensor with two indices, while a vector is a first-order ($n = 1$) tensor with a single index (see Figure 2.1). Tensors are fundamental objects in linear algebra and multilinear computations, enabling compact representations of high-dimensional data and operations on them. The elements of a tensor $\mathcal{T}$ are accessed via multi-indices, denoted $\mathcal{T}_{i_1,i_2,\ldots,i_n}$, where each i_k ranges from 1 to d_k, the size of dimension k. The shape of a tensor, given by the tuple $(d_1, d_2, \ldots, d_n)$, determines the total number of entries, which scales as the product of all dimension sizes. This exponential growth in storage and computation with dimension and order motivates the use of structured factorizations and tensor networks, which exploit sparsity and algebraic structure to make high-dimensional problems computationally tractable.

2.1.2 Tensor Contractions

Tensor contractions have emerged as one of the fundamental operations on tensors. At its core, a tensor contraction represents a generalized summation over shared indices between tensors. This operation constitutes many familiar linear algebra operations such as matrix multiplication, but extends naturally to higher-order interactions. Formally, a tensor contraction between two tensors $\mathcal{A}$ and $\mathcal{B}$ over a set of shared indices S is defined as:

$$\mathcal{C}_{j_1,\ldots,j_m,k_1,\ldots,k_n} = \sum_{i_1=1}^{d_1} \sum_{i_2=1}^{d_2} \cdots \sum_{i_r=1}^{d_r} \mathcal{A}_{i_1,\ldots,i_r,j_1,\ldots,j_m} \cdot \mathcal{B}_{i_1,\ldots,i_r,k_1,\ldots,k_n} \tag{2.1}$$

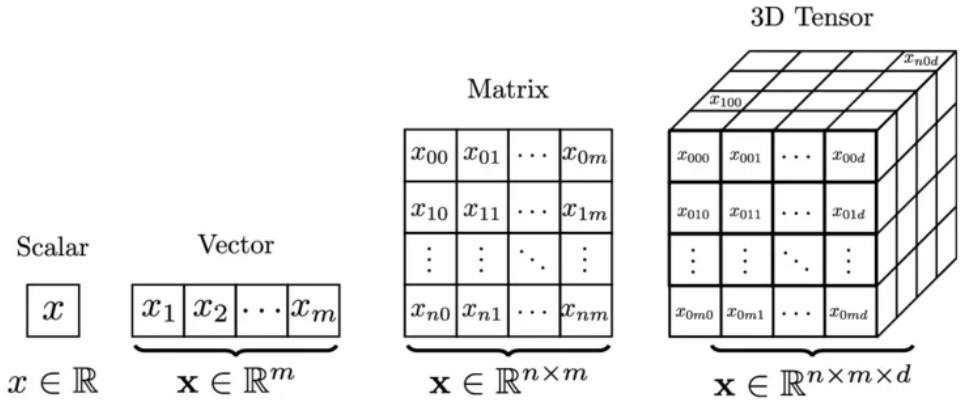

Figure 2.1: Examples of rank-0, 1, 2, 3 tensors (i.e. a scalar, vector, matrix, order-3 tensor)

where indices $i_1, \ldots, i_r$ are the contracted (summed) indices shared between $\mathcal{A}$ and $\mathcal{B}$, and indices $j_1, \ldots, j_m$ and $k_1, \ldots, k_n$ are the free indices that remain in the result.

A more compact and standard representation of tensor contractions is given by Einstein notation (Einstein summation convention), where repeated indices are implicitly summed over. For instance, the contraction above can be written concisely as:

$$\mathcal{C}_{jk} = \mathcal{A}_{ij}\mathcal{B}_{ik} \tag{2.2}$$

where i is the repeated (contracted) index and is implicitly summed over all its dimensions, while j and k are free indices appearing in the result. This notation extends naturally to higher-order tensors and multiple contractions, making it a powerful tool for expressing complex tensor network computations in a compact form.

2.1.3 Tensor Networks

However, this is not the only way to effectively represent complex contractions of networks; we can also draw them in tensor networks. A tensor network is a graphical representation of a set of tensors and their contraction structure, where nodes represent tensors and edges represent shared indices.

In this graphical framework, each tensor is depicted as a node (typically drawn as a circle, square, or other shape), with a number of lines (legs) extending from it equal to the tensor's order. Each leg corresponds to one index of the tensor. When two legs are connected by an edge, the corresponding indices are contracted (summed over). Free

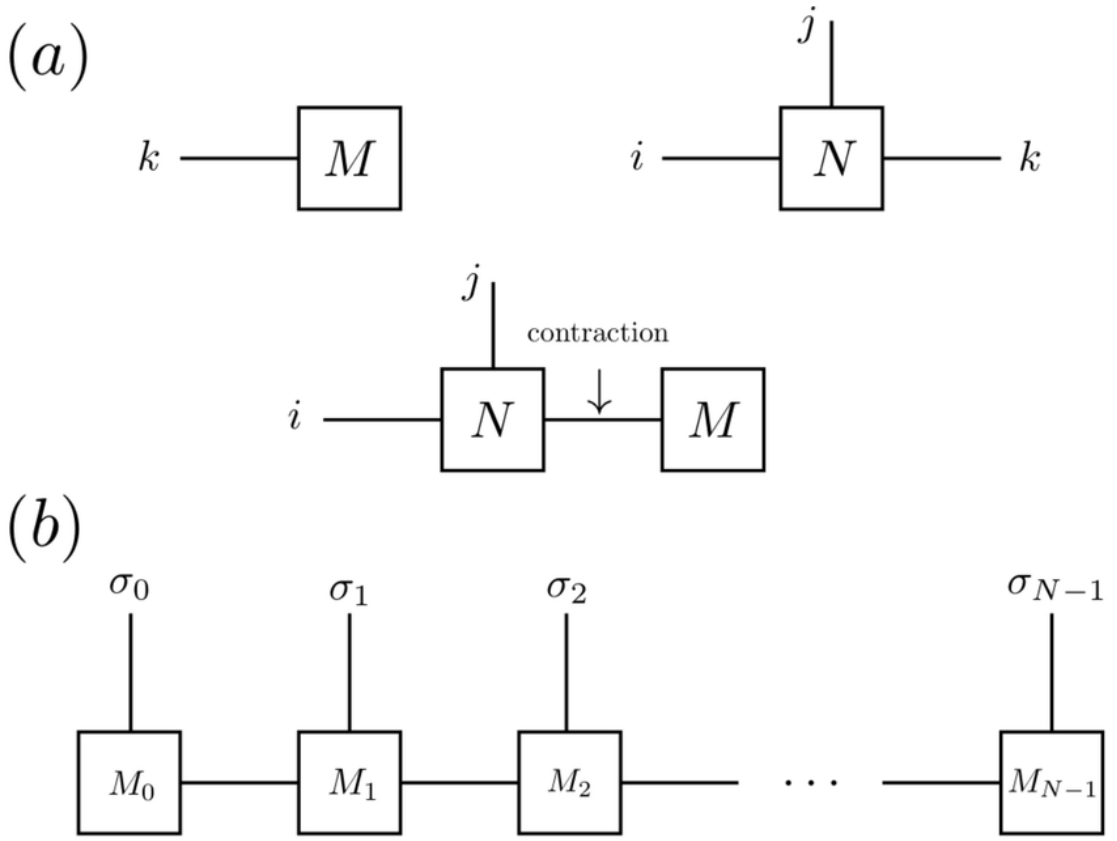

Figure 2.2: (a), on the top, displays a vector, M, and a rank-3 tensor, N, contracted across their common leg k; (b), on the bottom, displays a complex network of N tensors chained in a tensor train, commonly known as a Matrix Product State (MPS)

indices, which do not participate in contractions, are drawn as dangling legs extending from the network.

Consider a simple example: contracting two tensors $\mathcal{A}_{ij}$ and $\mathcal{B}_{jk}$ to produce $\mathcal{C}_{ik}$. In tensor network notation, $\mathcal{A}$ appears as a node with two legs (for indices i and j), and $\mathcal{B}$ appears as a separate node with two legs (for indices j and k). The leg corresponding to index j in $\mathcal{A}$ is connected to the leg corresponding to index j in $\mathcal{B}$, indicating that these indices are contracted. The remaining legs (corresponding to i and k) dangle freely, representing the free indices of the result tensor $\mathcal{C}$. Figure 2.2 illustrates this concept with more examples.

More complex tensor networks can compactly represent intricate contraction patterns

that would be cumbersome to express algebraically. The graphical representation immediately reveals the structure of the network i.e., which tensors share indices, the connectivity pattern, and which indices remain uncontracted. This makes tensor network diagrams important for visualizing and reasoning about large-scale contractions [3].

2.2 Basic Linear Algebra Subprograms

Basic Linear Algebra Subprograms (BLAS) is a standardized specification for low-level linear algebra operations that serves as a foundation for high-performance numerical computing. Originally developed in the late 1970s [14], BLAS provides a portable and efficient interface for fundamental vector and matrix operations, enabling libraries and applications to achieve near-optimal performance across diverse hardware architectures.

BLAS operations are organized into three hierarchical levels, each characterized by the computational intensity (ratio of arithmetic operations to memory accesses) and scope of the operations:

2.2.1 Level 1: Vector Operations

Level 1 BLAS conserns with scalar and vector operations with computational complexity $O(n)$, where n is the vector length. These operations include vector addition, scalar multiplication, dot products, and norm calculations. Due to their linear computational intensity and high memory-to-computation ratio, Level 1 operations are memory-bound and exhibit poor cache utilization. Examples include `AXPY` (scaled vector addition) (2.3), `DOT` (dot product), and `NRM2` (Euclidean norm).

$$y \leftarrow \alpha x + y \tag{2.3}$$

2.2.2 Level 2: Matrix-Vector Operations

Level 2 BLAS optimizes matrix-vector operations with computational complexity $O(n^2)$, including matrix-vector products, rank-one updates, and triangular system solves. These operations have moderate computational intensity and remain memory-bound on most modern architectures. Common Level 2 operations include `GEMV` (general matrix-vector multiply) (2.4) and `GER` (general rank-one update).

$$y \leftarrow \alpha A x + \beta y \tag{2.4}$$

2.2.3 Level 3: Matrix-Matrix Operations

Level 3 BLAS covers matrix-matrix operations with computational complexity $O(n^3)$, such as matrix multiplication and triangular solves. These operations exhibit high computational intensity and can achieve efficient cache reuse through careful data blocking and tiling strategies. Level 3 operations are typically compute-bound on modern hardware, making them suitable targets for optimization and parallelization. The general matrix-matrix multiplication (GEMM) (2.5) operation is the cornerstone of Level 3 BLAS.

$$C \leftarrow \alpha AB + \beta C \tag{2.5}$$

Efficient implementations of BLAS, such as ATLAS, OpenBLAS, and Intel MKL, are critical for achieving high performance in scientific computing and machine learning applications [8]. In the context of tensor contractions, BLAS operations, particularly Level 3 GEMM, serve as the primary computational kernel when contractions are transformed into matrix multiplications.

2.3 High-performance Matrix-Matrix Multiplication Implementations

The work presented uses the `Tensor Contraction Code Generator (TCCG)` [20] library as its high-performance matrix-matrix multiplication engine. `TCCG` is designed specifically to deliver efficient GEMM kernels across modern CPU architectures by combining the algorithmic principles of blocked matrix multiplication with architecture-aware microkernel generation.

At the highest level, `TCCG` decomposes the computation into a hierarchy of tiles. The input matrices are partitioned into blocks that fit into the different levels of the processor cache, which reduces main-memory traffic and improves data reuse. The blocked structure also enables the library to expose parallelism naturally: independent blocks can be computed concurrently across CPU cores, while each core operates on cache-resident panels of the matrices.

A central feature of `TCCG` is its use of a carefully tuned microkernel. The microkernel computes a small block of the result matrix using registers and vector instructions, often with explicit loop unrolling and packing of input matrix panels. This organization minimizes the number of memory accesses per floating-point operation, which is essential for reaching high fractions of peak performance on modern processors.

The library also supports explicit matrix packing, where subblocks of the input matrices are copied into contiguous, cache-friendly buffers before the inner product is

evaluated. Packing eliminates irregular memory access patterns that occur in standard row-major or column-major layouts, and it allows the microkernel to consume data in a streaming fashion. In TCCG, the packed panels are chosen to align with the target architecture's cache line size and vector width.

Beyond blocking and packing, TCCG implements an adaptive blocking strategy for common matrix shapes. For square matrices, the blocking factors are selected to balance between the L1 / L2 cache capacities and the cost of data movement. For rectangular matrices, the library selects block sizes to maintain high utilization of the inner microkernel while avoiding excessive padding or wasted computation.

It is important to emphasize that TCCG is not just treated as a black-box library, but integrated into the overall implementation. Its performance model is based on the same principles discussed earlier in this chapter: increasing arithmetic intensity, maximizing cache reuse, and exposing parallelism through block-level decomposition. When tensor contractions are mapped to matrix-matrix multiplications, TCCG provides the low-level kernel that performs these multiplications efficiently, making the overall tensor contraction implementation practical for large-scale problems.

2.4 High-performance Tensor Contraction Implementations

The main tensor transposition engine used in this work is the High-Performance Tensor Transposition (HPTT) library [22]. HPTT addresses a data-movement-centric bottleneck that is ubiquitous in tensor-based codes. While a single GEMM call can be made very fast, the full contraction pipeline often requires rearranging tensor data, and inefficient transpositions can dominate runtime if not handled carefully.

HPTT implements multidimensional, out-of-place tensor transpositions using the same high-performance principles introduced earlier for matrix multiplication: blocking, explicit vectorization, a small hand-tuned micro-kernel, and parallelization. Concretely, HPTT decomposes any higher-order permutation into a nest of loops surrounding a micro-kernel that performs a small, cache-resident element reordering. This decomposition has three practical advantages: (1) the micro-kernel can be hand-vectorized for the target instruction set (e.g., AVX, NEON), (2) blocking at multiple levels improves cache reuse and reduces off-chip bandwidth requirements, and (3) the loop nest exposes both data- and task-level parallelism for multicore execution.

$$B_{\Pi(i_1,i_2,\ldots,i_N)} \leftarrow \alpha \times A_{i_1,i_2,\ldots,i_N} + \beta \times B_{\Pi(i_1,i_2,\ldots,i_N)}$$

Figure 2.3: Illustration of HPTT's tensor transposition semantics and permutation notation (image source: springer13/hptt).

An important engineering feature of HPTT is its optional autotuning framework. At plan creation time, the library can either estimate good implementation parameters from a performance model (fast) or explore a search space of loop orders, blocking factors, and parallelization strategies (measure/patient modes) to find the best-performing schedule for the current tensor shape and hardware. This behavior is analogous to FFTW's planner [6] and makes HPTT robust in applications where tensor sizes and permutations are only known at runtime.

HPTT also supports explicit matrix/tensor packing and specializes micro-kernels to align with cache line sizes and vector register widths, thereby minimizing irregular memory accesses and enabling streaming consumption of input data. The library exposes simple C, C++, and Python interfaces for creating a transposition plan (with or without autotuning) and executing it, for example see Figure 2.4.

```
#include <hptt.h>

// allocate tensors
float A* = ...
float B* = ...

// specify permutation and size
int dim = 6;
int perm[dim] = {5,2,0,4,1,3};
int size[dim] = {48,28,48,28,28};

// create a plan (shared_ptr)
auto plan = hptt::create_plan( perm, dim,
                           alpha, A, size, NULL,
                           beta, B, NULL,
                           hptt::ESTIMATE, numThreads);

// execute the transposition
plan->execute();
```

Figure 2.4: Example usage of HPTT's C++ interface from the library's documentation.

In practice, integrating HPTT as the transposition layer in a tensor contraction pipeline significantly reduces the overall runtime, because the cost of rearranging tensor data becomes comparable, if not superior, to the cost of the high-performance GEMM kernels if not optimized.

3 Related Work

Early work primarily optimized for FLOPs, but it is now well understood that minimizing FLOPs alone can lead to prohibitively large intermediate tensors and thus excessive memory consumption. Therefore, a new set of algorithms have been developed to reduce this effect of the memory. This section reviews these contributions, with a focus on techniques that explicitly reduce memory footprint or data movement.

3.1 Treewidth-Based and Graph-Theoretic Methods

Markov and Shi [15] established a foundational connection between tensor network contraction and graph theory, showing that the complexity of contraction is governed by the treewidth of the underlying graph. Minimizing treewidth implicitly controls the size of intermediate tensors and therefore peak memory usage. This is due to the treewidth bounding the size of the largest intermediate tensor produced during contraction. A contraction with treewidth k guarantees that no intermediate tensor will have more than k indices. This directly translates to memory savings. Moreover, low treewidth enables efficient dynamic programming algorithms that can compute the optimal contraction sequence in time exponential only in treewidth, not in the total number of tensors.

However, exact treewidth minimization is NP-hard, motivating heuristic approaches. Subsequent work has leveraged graph partitioning and elimination ordering heuristics to approximate optimal contraction paths. For example, tools such as `quickbb` [7] and `flow-cutter` [11] have been adapted to tensor contraction problems. These approaches reduce peak memory by limiting separator sizes, though they are not explicitly designed for memory movement.

3.2 Hypergraph Partitioning Approaches

More recent work models tensor networks as hypergraphs, enabling more accurate representation of multi-index tensors. Gray and Kourtis [9] introduced a hypergraph-based contraction path optimizer that uses partitioning techniques to minimize both

FLOPs and intermediate tensor sizes. This framework allows explicit incorporation of memory constraints and has been widely adopted in tools such as `cotengra`.

Hypergraph partitioning methods can be extended to multi-objective optimization, balancing FLOPs and memory. For instance, multi-criteria cost functions penalize large intermediate tensors, effectively reducing peak memory usage. However, these methods typically focus on static path selection and do not account for runtime memory movement.

3.3 Multi-Objective and Memory-Aware Search

Recent advances have emphasized multi-objective optimization, where contraction paths are evaluated based on both computational cost and memory footprint. For example, Gray [10] introduced heuristic search strategies that explicitly track peak tensor size during path construction. Similarly, approaches based on simulated annealing or genetic algorithms incorporate memory penalties into the objective function.

Despite these advances, most pre-contraction methods treat memory as a scalar constraint (peak size) rather than addressing *memory movement* or data locality. As a result, they may produce paths that are optimal in terms of peak memory but inefficient in practice due to poor data reuse or excessive data transfers.

While pre-contraction methods determine the order of operations, post-contraction strategies modify how contractions are executed to reduce memory usage and movement. These techniques are particularly relevant when the contraction path is fixed or when dynamic adjustments are needed at runtime.

3.4 Slicing Techniques

Slicing is one of the most widely used methods for reducing peak memory. It involves fixing certain indices to specific values, thereby decomposing a large contraction into multiple smaller subproblems. Each slice can be computed independently, significantly reducing memory requirements at the cost of increased total computation.

This approach has been extensively used in quantum circuit simulation, including in Google's quantum supremacy experiments [1]. Later work formalized slicing as an optimization problem, selecting indices to slice in order to trade off memory and runtime [9]. Slicing effectively reduces peak memory but increases memory movement due to repeated loading and storing of intermediate data.

3.5 Recompression and Tensor Decomposition

Another class of methods reduces memory by compressing intermediate tensors. Techniques such as singular value decomposition (SVD) and tensor train (TT) decompositions are used to approximate tensors with lower-rank representations.

For example, methods based on matrix product states (MPS) [19] and tensor trains [16] dynamically truncate intermediate tensors to control memory growth. These approaches are particularly effective in structured problems with low entanglement, but introduce approximation error and may not generalize to arbitrary networks.

3.6 Out-of-Core and Memory Hierarchy Optimization

Out-of-core methods address memory limitations by storing intermediate tensors on disk or across distributed memory systems. These approaches aim to minimize data movement between memory levels, which is often the dominant cost in large-scale contractions.

Communication-avoiding algorithms [2] provide a theoretical framework for minimizing data movement in linear algebra operations, and similar principles have been applied to tensor contractions. For instance, distributed contraction frameworks partition tensors across nodes to reduce communication volume, though this introduces additional complexity in scheduling and synchronization.

3.7 Dynamic Reordering and Execution Scheduling

Some approaches modify the contraction order at runtime to adapt to memory constraints. Dynamic programming techniques can reorder operations based on available memory, while task-based runtimes schedule contractions to maximize data reuse.

These methods blur the line between pre- and post-contraction optimization, as they effectively refine the contraction path during execution. However, they remain relatively underexplored compared to static path optimization.

3.8 Memory Movement and Data Locality

While peak memory has received significant attention, memory movement is increasingly recognized as a critical bottleneck. Modern hardware architectures exhibit large disparities between computation speed and memory bandwidth, making data movement a dominant cost.

Efforts to reduce memory movement include blocking and tiling strategies, which aim to maximize cache reuse, as well as layout transformations that improve data locality. In the context of tensor contractions, permutation of tensor indices plays a key role in determining memory access patterns.

However, most existing work treats permutations as a secondary concern, focusing primarily on algebraic correctness rather than memory efficiency. This represents a significant gap in the literature.

3.9 Summary and Research Gap

In summary, existing work on memory reduction in tensor network contractions can be broadly categorized into pre-contraction path optimization and post-contraction execution strategies. Pre-contraction methods, particularly those based on hypergraph partitioning, have made significant progress in controlling peak memory. Post-contraction methods such as slicing and recompression provide additional flexibility but often increase computational cost or introduce approximation.

Despite these advances, several limitations remain:

- Most methods focus on peak memory rather than memory movement and data locality.
- Tensor permutations, which strongly influence memory access patterns, are not systematically optimized.
- There is limited integration between path optimization and low-level execution strategies.

The present work addresses these gaps by proposing a novel post-contraction optimization algorithm that analyzes tensor permutations to reduce both memory usage and memory movement. By focusing on execution-level transformations, this approach complements existing path optimization techniques and targets a critical, underexplored aspect of tensor network contraction.

Abbreviations

TNC Tensor network contraction

FLOPS floating-point operations

HPC high-performance computing

BLAS Basic Linear Algebra Subprograms

MPS Matrix Product State

GEMM general matrix-matrix multiplication

TCCG Tensor Contraction Code Generator

HPTT High-Performance Tensor Transposition

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